Matched Prime–Möbius Cutoff Shells: Rank–Two Factorization, Determinant–Frequency Dispersion, and the Distinguished Zero–Frequency Gate

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Abstract

We continue the primitive Möbius-tail reduction at the point where canonical determinant cells are individually sparse but their untwisted reassembly remains uncontrolled. Let $Q = X^{\beta+o(1)}$ be the row scale, $J = X^{1-\beta+o(1)}$ the orbit scale, and $N_0 = JQ^2 \asymp XQ$ the natural two-row mass. The two mixed ultra-cutoff channels and the both-ultra channel admit the pointwise identity

$$\mathsf{A}_{m,T}C_n + C_m\mathsf{A}_{n,T} + C_mC_n = \mathsf{A}_{m,U_0}\mathsf{A}_{n,U_0} - \mathsf{A}_{m,T}\mathsf{A}_{n,T}.$$

Consequently, their divisor coefficient is the difference of two rank-one tensors and has rank at most two. We retain this matching, the actual prime–Möbius row coefficients, the canonical target content, and the joint physical multiplier in one exact bilinear formula. The row sign and the reflected divisor sign fuse as

$$\mu(d)b_R(w) = -\mu(dw)w_R(w) \qquad \big(w \mid \ell dj + h\big),$$

while exact content and determinant phase factorize without requiring any coprimality between the content and the auxiliary Fourier modulus.

For the signed amplitude $A_C(n)$ of the matched shell with canonical content at most $C \le J$, $C \ll_{\mathscr{D}} Q$, and normalized determinant n, the preceding singleton-cell estimate gives

$$\|A_C\|_1 \ll X^\varepsilon N_0, \qquad \|A_C\|_\infty \ll X^\varepsilon Q(C+J), \qquad \|A_C\|_2^2 \ll X^\varepsilon \frac{N_0^2}{Q}.$$

Choosing a prime Fourier modulus larger than the determinant span yields an exact no-wrap Parseval identity. Hence all but $O(X^{2\sigma+\varepsilon})$ frequencies satisfy $|\widehat{A}_{C,q}(r)| \le N_0X^{-\sigma}$. A classical additive large sieve gives the corresponding Farey-family estimate. At $\beta = 267/400$, all but an $O(X^{-1/400+o(1)})$ proportion of determinant frequencies already have every fixed saving below $133/400$.

The remaining hard raw component of the original calibrated residual is the distinguished coefficient $r = 0$, not a generic frequency, modulo the already controlled large-content and drift terms. We show exactly that the nontrivial terms of the small-content Möbius projector reassemble to minus the previously bounded large-content shell, so the zero coefficient is equivalent, within the inherited large-content error, to the full raw cutoff shell. If its flatness exponent χ satisfies

$$|\widehat{A}_{C,q}(0)|^2 \le X^{\chi+o(1)} \sum_n |A_C(n)|^2,$$

then, whenever $\chi < \beta$ and $\kappa > 0$, the matched shell has every saving below

$$\min\left\{1-\beta,\, \kappa,\, \frac{\beta-\chi}{2}\right\}, \qquad C = X^{\kappa+o(1)}.$$

At the high-β endpoint with $C = \lfloor J \rfloor$, the precise remaining condition is $\chi \le 1/400$. This flatness estimate is not proved here. A sharp finite Fourier model shows that the available support and norm bounds, even combined with perfect cancellation at every nonzero frequency, cannot control $r = 0$. A primitive finite witness likewise shows that the content projector retains a coherent row mode. Thus the paper supplies an unconditional almost-all-frequency theorem, an exact conditional transfer, and a sharp interface obstruction; it does not prove a prime-pair asymptotic, twin primes, or a breach of the sieve parity barrier.

Keywords: primitive Möbius tail; matched cutoff shell; rank-two factorization; canonical target content; determinant frequency; large sieve; zero-frequency flatness; parity barrier.

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1 Introduction and route ledger

The primitive Möbius-tail programme has reached a sharply localized analytic question. The preceding reductions first preserved the four prime/model channels long enough to recombine their divisor coefficients, then returned from a selected cutoff to the full physical cutoff, and finally removed target pairs with large common content [7–11]. TPC-31 introduced the factorization-independent coordinates

$$N_m(j) = mj + h, \qquad c = (N_m(j), N_n(j)), \qquad \Delta^{\#} = \frac{m-n}{c}, \tag{1}$$

and proved fixed-power estimates for individual translated windows and residue cells of $\Delta^{\#}$ [12]. Absolute reassembly of all cells, however, returns the original shell. Equivalently, the untwisted sum is the zero coefficient of an auxiliary finite Fourier transform.

The purpose of this paper is to determine exactly how much of that Fourier problem follows from the available physical estimates, and exactly what new input is still missing. Three features must remain visible at the same time:

(i) the two mixed cutoff channels and the both-ultra channel have to be matched before absolute values are taken;

(ii) the outer Möbius sign on an opened row has to be fused with the reflected divisor sign;

(iii) the auxiliary determinant phase may be separated row by row, but the actual pair mask must remain a joint multiplier unless a controlled projective decomposition has been proved.

Keeping these three points produces a rank-two algebraic core and an exact determinant-frequency representation. The resulting mean-square theorem is strong enough on average, and at every frequency outside a power-small exceptional set. What remains is a single distinguished coefficient.

1.1 The unconditional results

Write

$$Q = X^{\beta+o(1)}, \qquad J = X^{1-\beta+o(1)}, \qquad QJ \asymp X, \qquad N_0 = JQ^2 \asymp XQ. \tag{2}$$

The inherited prefix $\mathsf{A}_{m,T}$ and ultra increment C_m obey the exact pointwise recombination

$$\mathsf{A}_{m,T}C_n + C_m\mathsf{A}_{n,T} + C_mC_n = \mathsf{A}_{m,U_0}\mathsf{A}_{n,U_0} - \mathsf{A}_{m,T}\mathsf{A}_{n,T}. \tag{3}$$

Thus the divisor-coefficient matrix is the difference of two rank-one tensors. This low rank is an exact matching statement, not a cancellation estimate: a rank-one tensor may carry the entire coherent mass.

There is also an exact arithmetic factorization. For every $C, q \geq 1$ and $r \pmod q$,

$$\begin{aligned}\mathbf{1}_{(N_m,N_n)\leq C}e_q\left(-r\frac{m-n}{(N_m,N_n)}\right)\\ = \sum_{c\leq C}\sum_{k\geq 1}\mu(k)\mathbf{1}_{ck|N_m}\mathbf{1}_{ck|N_n}e_{cq}(-r(m-n)).\end{aligned} \tag{4}$$

The sum is finite. Its phase splits as $e_{cq}(-rm)e_{cq}(rn)$, with no assumption that $(c, q) = 1$. Together with (3), this gives a factorable row core with the full joint physical multiplier retained.

Let $A_C(n)$ be the signed amplitude of the matched three-channel shell with canonical content at most C and normalized determinant n. The TPC-31 singleton-cell estimate and total absolute mass imply, for $C \leq J$ and $C \ll_{\mathscr{D}} Q$,

$$\|A_C\|_1 \ll X^\varepsilon N_0, \qquad \|A_C\|_\infty \ll X^\varepsilon Q(C+J), \qquad \|A_C\|_2^2 \ll X^\varepsilon \frac{N_0^2}{Q}. \tag{5}$$

A prime modulus larger than the determinant span eliminates residue collisions, so Parseval is an identity for the physical determinant amplitudes, not merely for grouped residue classes. It follows that

$$\#\{r \pmod q : |\widehat{A}_{C,q}(r)| > N_0X^{-\sigma}\} \ll X^{2\sigma+\varepsilon}. \tag{6}$$

The classical additive large sieve gives the analogous statement over reduced rational frequencies. No spectral large sieve or unproved Möbius hypothesis enters these conclusions.

1.2 The distinguished zero-frequency gate

Define

$$\lambda_C(t) = \sum_{\substack{c|t\\ c\leq C}} \mu(t/c). \tag{7}$$

Then

$$\mathbf{1}_{(N_m,N_n)\leq C} = \sum_{t|N_m,\ t|N_n} \lambda_C(t), \tag{8}$$

while $\lambda_C(1) = 1$ and, for $g = (N_m, N_n)$,

$$\sum_{\substack{t|g\\ t>1}} \lambda_C(t) = -\mathbf{1}_{g>C}. \tag{9}$$

Thus the unit-divisor term is the untwisted full-content shell, and every nonunit projector term together is exactly minus the large-content correction already treated in TPC-30. The cell decomposition does not center the coefficient needed by the original problem.

To quantify the remaining input, we introduce a zero-frequency flatness exponent χ through

$$|\widehat{A}_{C,q}(0)|^2 \leq X^{\chi+o(1)}\sum_n |A_C(n)|^2. \tag{10}$$

This is a hypothesis, not a conclusion of the paper. If $C = X^{\kappa+o(1)} \leq J$ and $C \ll_{\mathscr{D}} Q$, the exact transfer theorem gives every saving exponent strictly below

$$\min\left\{1-\beta,\ \kappa,\ \frac{\beta-\chi}{2}\right\}. \tag{11}$$

At

$$\beta = \frac{267}{400}, \qquad C = \lfloor J\rfloor = X^{133/400+o(1)}, \tag{12}$$

the precise threshold for recovering every saving below 133/400 is

$$\boxed{\chi \leq \frac{1}{400}.} \tag{13}$$

Unconditionally, all but an $X^{-1/400+o(1)}$ proportion of determinant frequencies already meet the target scale. The theorem needed by the remaining hard raw component of the original calibrated residual is the assertion that the special frequency $r = 0$ is not in that exceptional set, after the inherited large-content and drift terms are separated.

1.3 Why this is a genuine new gate

The interface obstruction is sharp. A constant signal on one complete residue system has the norm bounds in (5), has zero Fourier transform at every nonzero frequency, and nevertheless has maximal zero coefficient. Likewise, the small-content projector on pairwise coprime targets is the matrix $\mathbf{J} - I$; it retains the constant-row direction with eigenvalue of order the number of rows. Therefore support length, $\ell^1/\ell^\infty/\ell^2$ control, low rank, and nonzero-frequency cancellation cannot by themselves prove (13).

Opening one reflected ultra leg exposes the next arithmetic object. A divisibility condition parametrizes the row variable by one progression, and the two Möbius signs become a correlation of two nonproportional affine forms. Classical one-frequency Möbius estimates do not estimate that correlation, while known logarithmically averaged two-point results do not provide the uniform polynomial-scale physical estimate required here. This identifies a concrete next problem without asserting that an existing theorem closes it.

The conclusions of this paper are therefore deliberately asymmetric. We prove the matched factorization, the almost-all-frequency theorem, the exact zero-frequency transfer, and the sharp no-go models. We do not prove the flatness hypothesis, a full residual estimate, a Hardy–Littlewood main term, positivity of a prime-pair count, the twin-prime conjecture, or a breach of the sieve parity barrier [3].

1.4 Organization

The inherited packet and exact cutoff shell are recorded in section 2. Exact content factorization and the content projector are developed in section 3. The actual prime–Möbius sign and the joint-mask boundary are treated in section 4. Determinant dispersion and zero-frequency transfer occupy sections 5 and 6, followed by the high-β ledger. The two sharp interface models and the primitive finite witness appear in section 8. We finish with the next affine-Möbius gate and the exact certificate.

2 The physical packet and the matched cutoff shell

This section fixes the physical object before any determinant-frequency transform is taken. The order is important: the two mixed channels and the both-ultra channel come from one difference of quadratic prefixes and must first be recombined. The interface is inherited from TPC-25 and TPC-29–TPC-31 [9–12].

2.1 Primitive opened rows

Fix $h \in \mathbb{Z} \setminus \{0\}$, and put

$$H = \operatorname{rad} |h|. \tag{14}$$

Let X tend to infinity and fix $0 < \delta < 1/2$. In one opened dyadic packet the scales satisfy

$$R = X^{1/2-\delta+o(1)}, \qquad Q = LD, \qquad QJ \asymp_{\mathscr{D}} X, \tag{15}$$

with

$$D \leq R^{1/2+o(1)}, \qquad L > \max\{2R, 2|h|\}. \tag{16}$$

Equivalently, when exponent notation is useful, we write

$$Q = X^{\beta+o(1)}, \qquad J = X^{1-\beta+o(1)}, \qquad 0 < \beta < 1. \tag{17}$$

Here $\mathscr{D}$ denotes all fixed support, smoothness, residue, and positive-margin data.

Rows are indexed by

$$\alpha = (\ell_\alpha, d_\alpha), \qquad m_\alpha = \ell_\alpha d_\alpha \asymp_{\mathscr{D}} Q, \tag{18}$$

where $\ell_\alpha \in [L/2, 2L]$ is prime and $d_\alpha \in [D, 2D] \cap [1, R^{1/2+o(1)}]$ is squarefree with $(d_\alpha, H) = 1$. The size separation makes m_α determine α uniquely. The actual one-row coefficients have the prime–Möbius provenance

$$\gamma_\alpha^{(i)} = \mu(d_\alpha)(\log \ell_\alpha)\omega_D^{(i)}(d_\alpha)\psi_L^{(i)}(\ell_\alpha/L)\zeta_\alpha^{(i)}, \qquad |\zeta_\alpha^{(i)}| \ll_{\mathscr{D}} 1, \quad i = 1, 2. \tag{19}$$

The smooth dyadic families are fixed and bounded. In particular

$$\sum_\alpha |\gamma_\alpha^{(i)}| \ll_{\mathscr{D}} Q, \tag{20}$$

$$\sum_\alpha |\gamma_\alpha^{(i)}|^2 \ll_{\mathscr{D}} Q \log(2L), \qquad i = 1, 2. \tag{21}$$

For an orbit point j, set

$$N_\alpha(j) = m_\alpha j + h. \tag{22}$$

Every contributing row and orbit point obeys

$$(m_\alpha, H) = (j, H) = 1, \qquad c_{\mathscr{D}} X \leq N_\alpha(j) \leq C_{\mathscr{D}} X, \tag{23}$$

and the joint j-support contains $O_{\mathscr{D}}(J)$ integers. Consequently every divisor $w \mid N_\alpha(j)$ satisfies

$$(w, m_\alpha H) = 1. \tag{24}$$

The row-pair mask and the remaining physical weights are retained in the single joint multiplier

$$\mathfrak{A}_{\alpha,\gamma}(j) := \mathfrak{m}(\alpha, \gamma)\Xi_{\alpha,\gamma}(j)\mathcal{W}_{\alpha,\gamma}(j/J). \tag{25}$$

We use only

$$|\mathfrak{A}_{\alpha,\gamma}(j)| \ll_{\mathscr{D}} 1, \qquad \mathfrak{A}_{\alpha,\gamma}(j) = 0 \quad \text{if } m_\alpha = m_\gamma, \tag{26}$$

and that $\mathfrak{m}$ is independent of the divisor variables. The fixed-residue and smooth parts of (25) possess the finite Fourier and Mellin separations recorded in TPC-25. No separability of the generic row-pair mask is assumed here.

For a kernel $\mathcal{K}_{\alpha,\gamma}(j)$ and an event $\mathcal{E}(\alpha, \gamma, j)$, define

$$\mathcal{S}[\mathcal{K}; \mathcal{E}] := \sum_{\alpha,\gamma} \gamma_\alpha^{(1)}\gamma_\gamma^{(2)} \sum_j \mathfrak{A}_{\alpha,\gamma}(j)\mathcal{K}_{\alpha,\gamma}(j)\mathbf{1}_{\mathcal{E}(\alpha,\gamma,j)}. \tag{27}$$

The natural absolute scale is

$$JQ^2 \asymp XQ. \tag{28}$$

2.2 Divisor coefficients and exact prefixes

Put

$$a(w) = -\mu(w)\log w, \tag{29}$$

$$\lambda'_R(w) := \begin{cases} w \displaystyle\sum_{b\le R/w} \frac{\mu(wb)\mu(b)}{\varphi(wb)}, & w \le R, \\ 0, & w > R, \end{cases} \tag{30}$$

$$b_R(w) = a(w) - \lambda'_R(w). \tag{31}$$

All three coefficients vanish off the squarefree integers. On squarefree support,

$$b_R(w) = -\mu(w)w_R(w), \tag{32}$$

$$w_R(w) := \log w + \mathbf{1}_{w\le R}\frac{w}{\varphi(w)} \sum_{\substack{b\le R/w \\ (b,w)=1}} \frac{\mu(b)^2}{\varphi(b)} > 0. \tag{33}$$

Choose integers

$$R \le T < U_0, \qquad U_0 \asymp_{\mathscr{D}} X, \tag{34}$$

with U_0 larger than every target on (23). Define the raw prefix

$$\mathsf{A}_{m,Y}(j) := \sum_{\substack{w\le Y \\ w\mid mj+h}} b_R(w), \qquad R \le Y \le U_0, \tag{35}$$

and the ultra increment

$$C_m(j) := \sum_{\substack{T<w\le U_0 \\ w\mid mj+h}} a(w). \tag{36}$$

Since $T \ge R$, one has $b_R(w) = a(w)$ throughout the new shell. Thus, pointwise in m, j,

$$\boxed{\mathsf{A}_{m,U_0}(j) = \mathsf{A}_{m,T}(j) + C_m(j).} \tag{37}$$

The elementary divisor estimate gives, uniformly on the physical support,

$$|\mathsf{A}_{m,T}(j)| + |\mathsf{A}_{m,U_0}(j)| + |C_m(j)| \ll_{\varepsilon,\mathscr{D}} X^{\varepsilon}. \tag{38}$$

The actual row coefficient and the divisor coefficient fuse on each leg without an extra hypothesis.

Lemma 2.1 (Prime–Möbius sign fusion). *If $w \mid N_\alpha(j)$ and $b_R(w) \ne 0$, then*

$$\boxed{\mu(d_\alpha)b_R(w) = -\mu(d_\alpha w)w_R(w).} \tag{39}$$

Proof. The support of b_R makes w squarefree, while d_α is squarefree by construction. Since $d_\alpha \mid m_\alpha$, target primitivity (24) gives $(d_\alpha, w) = 1$. Hence $\mu(d_\alpha)\mu(w) = \mu(d_\alpha w)$, and (32) proves the claim. □

Remark 2.2 (A fusion which is not available). The Möbius factor introduced later by exact-content inversion may share prime factors with w. It therefore cannot be fused with $\mu(w)$ unless an additional coprimality condition has first been proved. No such fusion is used below.

2.3 Three raw channels as one rank–two shell

For two rows $m = m_\alpha$ and $n = m_\gamma$, define

$$\begin{aligned}\mathcal{K}^{(1)}_{\alpha,\gamma}(j) &= \mathsf{A}_{m,T}(j)C_n(j),\\ \mathcal{K}^{(2)}_{\alpha,\gamma}(j) &= C_m(j)\mathsf{A}_{n,T}(j),\\ \mathcal{K}^{(3)}_{\alpha,\gamma}(j) &= C_m(j)C_n(j).\end{aligned} \tag{40}$$

Their matched sum is

$$\mathcal{K}^{\mathrm{sh}}_{\alpha,\gamma}(j) := \sum_{\nu=1}^{3} \mathcal{K}^{(\nu)}_{\alpha,\gamma}(j). \tag{41}$$

Proposition 2.3 (Exact cutoff shell)**.** *Pointwise in every physical row pair and orbit point,*

$$\boxed{\mathcal{K}^{\mathrm{sh}}_{\alpha,\gamma}(j) = \mathsf{A}_{m,U_0}(j)\mathsf{A}_{n,U_0}(j) - \mathsf{A}_{m,T}(j)\mathsf{A}_{n,T}(j).} \tag{42}$$

Equivalently, after opening both divisor sums,

$$\mathcal{K}^{\mathrm{sh}}_{\alpha,\gamma}(j) = \sum_{u,v\geq 1} b_R(u)b_R(v)\Theta_{T,U_0}(u,v)\mathbf{1}_{u|N_\alpha(j)}\mathbf{1}_{v|N_\gamma(j)}, \tag{43}$$

$$\Theta_{T,U_0}(u,v) := \mathbf{1}_{u\leq U_0}\mathbf{1}_{v\leq U_0} - \mathbf{1}_{u\leq T}\mathbf{1}_{v\leq T}. \tag{44}$$

In particular Θ_{T,U_0}, viewed as a matrix in u, v, has algebraic rank at most two.

Proof. Insert (37) on both rows and expand the difference of products. This gives the three terms in (40). Opening the prefixes proves (43). Finally, with $p_Y(u) = \mathbf{1}_{u\leq Y}$,

$$\Theta_{T,U_0} = p_{U_0} \otimes p_{U_0} - p_T \otimes p_T,$$

a difference of two rank-one tensors. □

There is a useful symmetric polarization which keeps the same rank bound. Put

$$d_{T,U_0} = p_{U_0} - p_T, \qquad s_{T,U_0} = \frac{p_{U_0} + p_T}{2}. \tag{45}$$

Then

$$\boxed{\Theta_{T,U_0} = d_{T,U_0} \otimes s_{T,U_0} + s_{T,U_0} \otimes d_{T,U_0}.} \tag{46}$$

At the level of the physical divisor sums, define

$$\mathsf{H}_{m;T,U_0}(j) := \frac{\mathsf{A}_{m,U_0}(j) + \mathsf{A}_{m,T}(j)}{2} = \mathsf{A}_{m,T}(j) + \frac{1}{2}C_m(j). \tag{47}$$

Then the exact polarized shell is

$$\boxed{\mathcal{K}^{\mathrm{sh}}_{\alpha,\gamma}(j) = C_m(j)\mathsf{H}_{n;T,U_0}(j) + \mathsf{H}_{m;T,U_0}(j)C_n(j).} \tag{48}$$

Thus three rectangles become two polarized tensors before estimation. This is an algebraic rank statement. It does not make the masked row form positive semidefinite and does not authorize deletion of either polarized term.

2.4 The complete calibrated difference and its drift boundary

For $m = \ell d$, let

$$\delta_H(m) = \frac{H}{\varphi(H)} \frac{d}{\varphi(d)(\ell-1)} \ll_{\varepsilon,h} \frac{X^\varepsilon}{L}, \tag{49}$$

and define the calibrated prefix

$$P_{m,Y}(j) = \mathsf{A}_{m,Y}(j) - \delta_H(m). \tag{50}$$

Proposition 2.4 (Exact calibrated cutoff difference)**.** *For every physical m, n, j,*

$$\begin{aligned} &P_{m,U_0}(j)P_{n,U_0}(j) - P_{m,T}(j)P_{n,T}(j) \\ &\quad = \mathcal{K}^{\mathrm{sh}}_{\alpha,\gamma}(j) - \delta_H(m)C_n(j) - \delta_H(n)C_m(j). \end{aligned} \tag{51}$$

There is no two-drift term: it cancels between the two cutoff levels.

Proof. Use $P_{m,U_0} = P_{m,T} + C_m$, which follows from (37), and expand. Alternatively, insert (50) into the difference in (42); the two copies of $\delta_H(m)\delta_H(n)$ cancel. □

Let $\mathcal{D}^{\mathrm{dr}}(\mathcal{E})$ be the contribution of the final two terms in (51) to (27), with any additional event $\mathcal{E}$. Since an indicator or a unit-modulus determinant phase can only decrease its absolute majorant, the following estimate is uniform over every restriction used later.

Proposition 2.5 (Uniform drift boundary)**.** *For every fixed $\varepsilon > 0$,*

$$\boxed{|\mathcal{D}^{\mathrm{dr}}(\mathcal{E})| \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon \frac{XQ}{L}.} \tag{52}$$

Consequently, if $L \geq X^{\lambda_0}$ for fixed $\lambda_0 > 0$, the drift legs have a fixed-power saving relative to XQ, after choosing the soft-loss exponent below λ_0.

Proof. By (20) and (49),

$$\sum_\alpha |\gamma^{(i)}_\alpha| \delta_H(m_\alpha) \ll X^\varepsilon Q/L.$$

Use this on the drift row, (20) on the other row, (38) for the ultra increment, and the $O_{\mathscr{D}}(J)$ orbit length. Each of the two legs is

$$\ll X^\varepsilon J(Q/L)Q \asymp X^\varepsilon XQ/L.$$

Renaming the harmless soft-loss exponent proves the claim. □

Remark 2.6 (What is carried into the next sections)*.* The hard raw object is the single matched kernel (42), not three unrelated absolute sums. The calibrated packet equals that object plus the rigorously bounded drift contribution (52). Every factorization below retains the joint multiplier (25); boundedness alone is not treated as a factorization theorem.

3 Canonical-content inversion and the retained bilinear form

We now place the canonical determinant phase inside the exact cutoff shell. The central point is elementary but useful: exact target content can be imposed by a common-divisor Möbius projector, and the normalized determinant character then splits into one character on each row. The fixed residue factor, physical weight, and pair mask are not discarded in this operation.

3.1 Primitive common divisors

For two physical rows $m = m_\alpha$ and $n = m_\gamma$, abbreviate

$$N_m = mj + h, \qquad N_n = nj + h, \qquad G_{m,n}(j) = (N_m, N_n). \tag{53}$$

When $m \neq n$, the canonical normalized determinant of TPC-31 is

$$\Delta^{\#}_{m,n}(j) = \frac{m-n}{G_{m,n}(j)}. \tag{54}$$

Lemma 3.1 (Every common target divisor divides the row difference)**.** *Under* (23)*, if*

$$t \mid N_m, \qquad t \mid N_n, \tag{55}$$

then

$$(t, mnhj) = 1, \qquad t \mid m-n. \tag{56}$$

In particular

$$G_{m,n}(j) \mid m-n, \tag{57}$$

so (54) *is a nonzero integer.*

Proof. Target primitivity (24) applied on each row gives $(t, mnH) = 1$. If a prime p divided both t and j, then $p \mid N_m - mj = h$, hence $p \mid H$, a contradiction. Thus $(t, j) = 1$. Finally

$$t \mid N_m - N_n = (m-n)j,$$

and j may be cancelled modulo t. Taking $t = G_{m,n}(j)$ proves the last assertion. This is the primitive target-gcd geometry used in TPC-30 and TPC-31 [11, 12]. □

3.2 Exact-content Möbius projectors

The following formulation is convenient for both cumulative and dyadic content ranges.

Definition 3.2 (Content projector)**.** For a finite set $\mathscr{C} \subset \mathbb{N}$, define

$$\eta_{\mathscr{C}}(t) := \sum_{\substack{c \mid t \\ c \in \mathscr{C}}} \mu(t/c). \tag{58}$$

Proposition 3.3 (Exact gcd Möbius inversion)**.** *For all positive integers N_1, N_2, with $G = (N_1, N_2)$,*

$$\boxed{\mathbf{1}_{G \in \mathscr{C}} = \sum_{\substack{t \mid N_1 \\ t \mid N_2}} \eta_{\mathscr{C}}(t).} \tag{59}$$

Equivalently,

$$\mathbf{1}_{G=c} = \sum_{k \geq 1} \mu(k) \mathbf{1}_{ck \mid N_1} \mathbf{1}_{ck \mid N_2} \qquad (c \geq 1). \tag{60}$$

All sums in these identities are finite.

Proof. Interchanging the divisor sums gives

$$\sum_{t \mid G} \eta_{\mathscr{C}}(t) = \sum_{\substack{c \mid G \\ c \in \mathscr{C}}} \sum_{k \mid G/c} \mu(k).$$

The inner sum is one exactly when $G/c = 1$, and is zero otherwise. This proves (59). Taking $\mathscr{C} = \{c\}$ gives (60). □

Remark 3.4 (Size and sign). The coefficient $\eta_{\mathscr{C}}$ is signed and need not be multiplicative. The general pointwise estimate

$$|\eta_{\mathscr{C}}(t)| \le \tau(t) \tag{61}$$

does not by itself give cancellation after summation over t. The value of the projector is its exact reassembly, not an individual smallness property of its coefficients.

3.3 Canonical determinant phases

Write

$$e_q(x) = \exp(2\pi i x/q), \qquad q \ge 1. \tag{62}$$

Theorem 3.5 (Exact canonical phase splitting). *Let $q \ge 1$, $r \in \mathbb{Z}$, let $m \ne n$, and let $\mathscr{C} \subset \mathbb{N}$ be finite. Under the hypotheses of lemma 3.1,*

$$\begin{aligned} &\mathbf{1}_{G_{m,n}(j)\in\mathscr{C}}\, e_q\Big(-r\Delta^{\#}_{m,n}(j)\Big) \\ &\quad = \sum_{c\in\mathscr{C}}\sum_{k\ge 1} \mu(k)\mathbf{1}_{ck|N_m}\mathbf{1}_{ck|N_n} e_{cq}(-r(m-n)). \end{aligned} \tag{63}$$

For every summand on the right,

$$\boxed{e_{cq}(-r(m-n)) = e_{cq}(-rm)e_{cq}(rn).} \tag{64}$$

Neither identity requires $(c, q) = 1$.

Proof. Fix $c \in \mathscr{C}$. If ck divides both targets, then lemma 3.1 gives $c \mid m-n$, so every displayed character is well defined. For the actual content $G = G_{m,n}(j)$, the contribution of this c is zero unless $c \mid G$; when $c \mid G$, it equals

$$e_{cq}(-r(m-n)) \sum_{k|G/c} \mu(k).$$

Möbius inversion kills it unless $G = c$. In the surviving case,

$$e_{cq}(-r(m-n)) = e_q\left(-r\frac{m-n}{G}\right),$$

which proves (63). Equation (64) is the additive-character identity obtained by expanding $m-n$. No modular inverse of c has been used, so no coprimality between c and q is present. □

Remark 3.6 (Why the denominator is cq). It would be incorrect to replace division by c with multiplication by an inverse of $c \pmod q$. The phase is lifted from modulus q to modulus cq; this is precisely what makes (64) valid even when $(c, q) > 1$.

3.4 The complete retained-multiplier factorization

For a finite content family $\mathscr{C}$, define the matched determinant transform

$$\begin{aligned} \widehat{\mathcal{S}}^{\mathrm{sh}}_{\mathscr{C},q}(r) := \sum_{\substack{\alpha,\gamma \\ m_\alpha \ne m_\gamma}} \gamma^{(1)}_\alpha \gamma^{(2)}_\gamma \sum_j \mathfrak{A}_{\alpha,\gamma}(j)\mathcal{K}^{\mathrm{sh}}_{\alpha,\gamma}(j) \\ \times \mathbf{1}_{G_{m_\alpha,m_\gamma}(j)\in\mathscr{C}}\, e_q\Big(-r\Delta^{\#}_{m_\alpha,m_\gamma}(j)\Big). \end{aligned} \tag{65}$$

This is the sum of the transforms of all three raw channels, before any triangle inequality.

Put

$$\epsilon_{U_0} = 1, \qquad \epsilon_T = -1. \tag{66}$$

For $Y \in \{U_0, T\}$, define the two one-row factors

$$X^{(1)}_{\alpha;c,k,Y,r}(j) := \gamma^{(1)}_{\alpha} \mathbf{1}_{ck|N_\alpha(j)} \mathsf{A}_{m_\alpha,Y}(j) e_{cq}(-rm_\alpha), \tag{67}$$
$$X^{(2)}_{\gamma;c,k,Y,r}(j) := \gamma^{(2)}_{\gamma} \mathbf{1}_{ck|N_\gamma(j)} \mathsf{A}_{m_\gamma,Y}(j) e_{cq}(rm_\gamma). \tag{68}$$

Theorem 3.7 (Matched bilinear factorization with joint multiplier)**.** *Under the physical hypotheses of section 2, for every finite $\mathscr{C} \subset \mathbb{N}$, every $q \geq 1$, and every $r \in \mathbb{Z}$,*

$$\begin{aligned}\widehat{\mathcal{S}}^{\mathrm{sh}}_{\mathscr{C},q}(r) = \sum_{c\in\mathscr{C}} \sum_{k\geq 1} \mu(k) \sum_{j} \sum_{Y\in\{U_0,T\}} \epsilon_Y \\ \times \sum_{\substack{\alpha,\gamma \\ m_\alpha \neq m_\gamma}} \mathfrak{A}_{\alpha,\gamma}(j) X^{(1)}_{\alpha;c,k,Y,r}(j) X^{(2)}_{\gamma;c,k,Y,r}(j).\end{aligned} \tag{69}$$

The sums over k are finite. Formula (69) retains the exact rank–two cutoff difference and the complete joint physical multiplier.

Proof. Insert the cutoff difference (42) into (65). Apply theorem 3.5 to the content indicator and phase. The two row characters in (64), the two common-divisibility indicators, the two row coefficients, and the two prefixes are exactly the factors in (67)– (68). The joint multiplier is never altered. Since ck divides two targets of size $O_{\mathscr{D}}(X)$, one has $ck \ll_{\mathscr{D}} X$, proving finiteness. □

For fixed c, k, Y, r, j, let $M_j = (\mathfrak{A}_{\alpha,\gamma}(j))_{\alpha,\gamma}$. The innermost sum in (69) is the bilinear form

$$(X^{(1)}_{c,k,Y,r}(j))^{\mathsf{T}} M_j X^{(2)}_{c,k,Y,r}(j). \tag{70}$$

No complex conjugation is implicit. This notation makes the remaining boundary explicit: entrywise boundedness of M_j does not imply a bounded operator norm or a low-cost projective decomposition.

Corollary 3.8 (Conditional full product representation)**.** *Suppose, on a particular physical packet, that the retained multiplier has an explicit representation*

$$\mathfrak{A}_{\alpha,\gamma}(j) = \int_\Omega p_{\omega,j}(\alpha) q_{\omega,j}(\gamma)\, d\nu(\omega), \tag{71}$$

with absolute convergence sufficient to interchange all finite sums. Then the row-pair sum in (69) *equals*

$$\begin{aligned}\int_\Omega \left(\sum_\alpha p_{\omega,j}(\alpha) X^{(1)}_{\alpha;c,k,Y,r}(j) \right) \\ \times \left(\sum_\gamma q_{\omega,j}(\gamma) X^{(2)}_{\gamma;c,k,Y,r}(j) \right) d\nu(\omega).\end{aligned} \tag{72}$$

Proof. Substitute (71) and interchange the absolutely convergent integral and sums. □

The fixed-residue and physical smooth factors have such bounded-cost representations in the inherited packet [9]. The generic pair mask has not been proved to satisfy one with a sufficiently small projective norm. Thus corollary 3.8 is an exact conditional interface, not an estimate asserted from (26).

3.5 Dyadic projectors and exact zero-frequency reassembly

For $C \geq 1$, define the cumulative and dyadic coefficients

$$\eta_{\leq C}(t) := \sum_{\substack{c|t \\ c\leq C}} \mu(t/c), \tag{73}$$

$$\eta_{(C,2C]}(t) := \sum_{\substack{c|t \\ C<c\leq 2C}} \mu(t/c). \tag{74}$$

To match the cumulative notation used elsewhere in the paper, we also write

$$\lambda_C(t) := \eta_{\leq C}(t). \tag{75}$$

Lemma 3.9 (The first dyadic band)**.** *For every $C \geq 1$,*

$$\boxed{\lambda_C(1) = 1, \qquad \lambda_C(t) = 0 \ (2 \leq t \leq C), \qquad \lambda_C(t) = -1 \ (C < t \leq 2C).} \tag{76}$$

Proof. The first equality is immediate. If $2 \leq t \leq C$, every divisor of t is included in the definition, so

$$\lambda_C(t) = \sum_{c|t} \mu(t/c) = \sum_{d|t} \mu(d) = 0.$$

If $C < t \leq 2C$, every proper divisor of t is at most $t/2 \leq C$, whereas $c = t$ is omitted. Therefore

$$\lambda_C(t) = \sum_{\substack{c|t \\ c<t}} \mu(t/c) = \sum_{\substack{d|t \\ d>1}} \mu(d) = -\mu(1) = -1.$$

□

Corollary 3.10 (Exact cumulative and dyadic content projectors)**.** *For $G = (N_1, N_2)$,*

$$\mathbf{1}_{G\leq C} = \sum_{t|N_1,\, t|N_2} \eta_{\leq C}(t), \tag{77}$$

$$\mathbf{1}_{C<G\leq 2C} = \sum_{t|N_1,\, t|N_2} \eta_{(C,2C]}(t). \tag{78}$$

Moreover,

$$\eta_{(C,2C]}(t) = \eta_{\leq 2C}(t) - \eta_{\leq C}(t). \tag{79}$$

Hence cumulative projectors telescope exactly into dyadic content layers, with an optionally truncated final layer.

Proof. Apply proposition 3.3 to $\mathscr{C} = [1, C] \cap \mathbb{N}$ and $\mathscr{C} = (C, 2C] \cap \mathbb{N}$. The coefficient identity follows directly from the definitions. □

At determinant frequency zero, the exact-content labels may be reassembled into one common-divisor coefficient. Indeed, setting $r = 0$ in (63) and writing $t = ck$ gives

$$\mathbf{1}_{G\leq C} = \sum_{t|N_1,\, t|N_2} \eta_{\leq C}(t). \tag{80}$$

The term $t = 1$ is distinguished because

$$\eta_{\leq C}(1) = 1. \tag{81}$$

Proposition 3.11 ($t = 1$ and large-content reassembly). *For every positive integer G and every $C \geq 1$,*

$$\boxed{\sum_{\substack{t|G\\t>1}} \eta_{\leq C}(t) = -\mathbf{1}_{G>C}.} \tag{82}$$

Thus the $t = 1$ term of the cumulative projector is the unrestricted amplitude, while all $t > 1$ terms reassemble exactly to the negative large-content amplitude.

Proof. By (77), $\sum_{t|G} \eta_{\leq C}(t) = \mathbf{1}_{G\leq C}$. Subtract the $t = 1$ value (81) and use $\mathbf{1}_{G\leq C} - 1 = -\mathbf{1}_{G>C}$. □

To state the consequence for the physical shell, let

$$\mathcal{S}^{\mathrm{sh}}_{\mathrm{full}} := \mathcal{S}[\mathcal{K}^{\mathrm{sh}}; \mathrm{true}], \tag{83}$$

$$\mathcal{S}^{\mathrm{sh}}_{>C} := \mathcal{S}[\mathcal{K}^{\mathrm{sh}}; G_{m_\alpha,m_\gamma}(j) > C]. \tag{84}$$

Corollary 3.12 (Exact auxiliary-zero reassembly). *For every auxiliary modulus $q \geq 1$,*

$$\boxed{\widehat{\mathcal{S}}^{\mathrm{sh}}_{[1,C],q}(0) = \mathcal{S}^{\mathrm{sh}}_{\mathrm{full}} - \mathcal{S}^{\mathrm{sh}}_{>C}.} \tag{85}$$

More precisely, the $t = 1$ term in the common-divisor expansion is $\mathcal{S}^{\mathrm{sh}}_{\mathrm{full}}$, and the sum of all $t > 1$ terms is $-\mathcal{S}^{\mathrm{sh}}_{>C}$.

Proof. At $r = 0$, the phase in (65) is one. Apply proposition 3.11 pointwise to each physical row–orbit triple, then sum with the original signed coefficient and the retained multiplier. □

TPC-30 proves, uniformly for the sum of all three raw channels,

$$|\mathcal{S}^{\mathrm{sh}}_{>C}| \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon \left(Q^2 + \frac{XQ}{C}\right) \ll_{\mathscr{D}} X^\varepsilon XQ \left(\frac{1}{J} + \frac{1}{C}\right). \tag{86}$$

Combining this bound with (85) gives

$$\left|\widehat{\mathcal{S}}^{\mathrm{sh}}_{[1,C],q}(0) - \mathcal{S}^{\mathrm{sh}}_{\mathrm{full}}\right| \ll X^\varepsilon XQ \left(\frac{1}{J} + \frac{1}{C}\right). \tag{87}$$

This is an equivalence up to an already controlled large-content error, not an estimate for the unrestricted shell.

For the complete calibrated cutoff difference, the drift estimate (52) yields the corresponding boundary

$$\mathcal{K}^{\mathrm{cal}}_{\alpha,\gamma}(j) := P_{m_\alpha,U_0}(j)P_{m_\gamma,U_0}(j) - P_{m_\alpha,T}(j)P_{m_\gamma,T}(j), \tag{88}$$

$$\mathcal{S}^{\mathrm{cal}}_{\mathrm{full}} := \mathcal{S}[\mathcal{K}^{\mathrm{cal}}; \mathrm{true}],$$

$$\widehat{\mathcal{S}}^{\mathrm{cal}}_{[1,C],q}(0) := \mathcal{S}[\mathcal{K}^{\mathrm{cal}}; G_{m_\alpha,m_\gamma}(j) \leq C]. \tag{89}$$

The notation retains q to emphasize that this is the zero coefficient of the same auxiliary residue-label transform; its value is independent of q. One then has

$$\boxed{\left|\widehat{\mathcal{S}}^{\mathrm{cal}}_{[1,C],q}(0) - \mathcal{S}^{\mathrm{cal}}_{\mathrm{full}}\right| \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon XQ \left(\frac{1}{J} + \frac{1}{C} + \frac{1}{L}\right).} \tag{90}$$

Here both calibrated quantities are defined using the two-prefix difference in (51); the L^{-1} term accounts for the two drift legs uniformly under either the small- or large-content restriction.

Remark 3.13 (The zero-frequency gate is not removed). Equation (87) shows that controlling the auxiliary residue-label zero coefficient is, within the inherited large-content margin, the same problem as controlling the unrestricted matched shell. The exact Möbius projector exposes this fact but does not make the $t = 1$ contribution oscillatory. Bounds for nonzero determinant frequencies, or termwise absolute bounds for $\eta_{\leq C}(t)$, do not imply the missing estimate.

4 Prime–Möbius leg fusion and the joint-mask boundary

The factorization of the cutoff shell becomes useful only if the actual opened-row coefficient is retained. We now restore that provenance and separate what is algebraically exact from what would require a new norm estimate.

4.1 The physical row coefficient

For $i \in \{1, 2\}$, a row is $\alpha = (\ell, d)$ with

$$\ell \asymp L \text{ prime}, \qquad d \asymp D \text{ squarefree}, \qquad (d, H) = 1, \qquad m_\alpha = \ell d \asymp Q, \tag{91}$$

and has coefficient

$$\gamma_\alpha^{(i)} = \mu(d)(\log \ell)\omega_D^{(i)}(d)\psi_L^{(i)}(\ell/L)\zeta_\alpha^{(i)}. \tag{92}$$

Here the $\omega_D^{(i)}$ and $\psi_L^{(i)}$ range over fixed bounded smooth families and $\zeta_\alpha^{(i)}$ contains fixed-period characters or phases. The inherited estimates are

$$\sum_\alpha |\gamma_\alpha^{(i)}| \ll Q, \qquad \sum_\alpha |\gamma_\alpha^{(i)}|^2 \ll Q \log(2L). \tag{93}$$

These bounds use Chebyshev's estimate for the prime source and are stronger than replacing every logarithm by X^ε [9].

Recall the squarefree divisor coefficient

$$b_R(w) = -\mu(w)w_R(w), \tag{94}$$

where

$$w_R(w) = \log w + \mathbf{1}_{w \le R} \frac{w}{\varphi(w)} \sum_{\substack{a \le R/w \\ (a,w)=1}} \frac{\mu(a)^2}{\varphi(a)} \tag{95}$$

on squarefree w; both sides of (94) vanish off the inherited squarefree support. For $w > T \ge R$ this reduces to

$$b_R(w) = a(w) = -\mu(w) \log w. \tag{96}$$

Lemma 4.1 (Primitive one-leg sign fusion)**.** *Suppose that $w \mid m_\alpha j + h$ on the primitive physical support. Then $(w, d) = 1$, and*

$$\boxed{\gamma_\alpha^{(i)} b_R(w) = -\mu(dw)(\log \ell)\omega_D^{(i)}(d)\psi_L^{(i)}(\ell/L)\zeta_\alpha^{(i)} w_R(w).} \tag{97}$$

In the ultra range $w > T$,

$$\mu(d)a(w) = -\mu(dw) \log w. \tag{98}$$

Proof. If a prime p divided both w and d, then it would divide $m_\alpha = \ell d$ and $m_\alpha j + h$, hence $p \mid h$. This contradicts $(d, H) = 1$, because $H = \operatorname{rad}|h|$. Thus $(w, d) = 1$. The divisor coefficient is squarefree-supported, and d is squarefree, so multiplicativity gives $\mu(d)\mu(w) = \mu(dw)$. Substitute (94) into (92); (98) follows from (96). □

One may consequently write a cutoff row signal schematically as

$$\mathcal{T}_W^{(i)}(j) = -\sum_{\ell,d} \sum_{\substack{w \le W \\ w \mid \ell d j + h}} \mu(dw)(\log \ell)w_R(w)\omega_D^{(i)}(d)\psi_L^{(i)}(\ell/L)\zeta_{\ell,d}^{(i)}\Phi_{\ell,d}^{(i)}(j)$$
$$+ \text{ the calibrated drift}, \tag{99}$$

where Φ retains the actual fixed-residue and smooth physical factors. Changing variables to $n = dw$ makes the outer sign a single $\mu(n)$, but the inner coefficient still contains the factorization $n = dw$, the prime source ℓ, and the congruence $w \mid \ell dj + h$. Thus (99) is a prime–Möbius normal form, not a claim that the resulting coefficient is well-factorable in the technical sieve-theoretic sense.

4.2 What the smooth and residue factors permit

Fixed-period masks have a finite Fourier expansion of bounded total mass. The smooth factors have an absolutely convergent Mellin projective representation; after the actual row coefficients have been included, its weighted projective ℓ^1 mass is $O(Q^2)$ and its energy mass is $O(Q\log(2L))$ [9]. Consequently these specific factors may be separated and reassembled without introducing an X^ε projective loss.

The generic row-pair mask is different. Write the full joint multiplier as

$$\mathfrak{A}_{\alpha,\gamma}(j) = \mathfrak{m}(\alpha,\gamma)\Xi_{\alpha,\gamma}(j)\mathcal{W}_{\alpha,\gamma}(j/J). \tag{100}$$

The inherited abstract interface assumes only

$$|\mathfrak{m}(\alpha,\gamma)| \le 1, \qquad \mathfrak{m}(\alpha,\gamma) = 0 \quad \text{when } m_\alpha = m_\gamma. \tag{101}$$

Entrywise multiplication by such a matrix need not preserve positivity, rank, or spectral norm. Accordingly, the exact formula in section 3 is a factorable row core with (25) retained; it is not an unqualified product of two row sums.

Definition 4.2 (Controlled projective multiplier)**.** We say that $\mathfrak{A}$ has projective cost $\mathfrak{P}$ on a fixed physical packet if it admits a representation

$$\mathfrak{A}_{\alpha,\gamma}(j) = \int_\Omega p_{\omega,j}(\alpha)q_{\omega,j}(\gamma)\,d\nu(\omega) \tag{102}$$

whose required row norms and orbit seminorms integrate to at most $\mathfrak{P}$.

Under (102), each bilinear core may be written as an integral of products of row sums and any estimate uniform in the displayed seminorms may be reassembled with cost $\mathfrak{P}$. Finite-dimensional singular-value decomposition alone does not give the needed theorem: its projective norm may be as large as the obstacle scale. No controlled decomposition for the generic pair mask is asserted here.

Remark 4.3 (Two different factorization statements). The cutoff coefficient has algebraic rank at most two in the divisor variables. The pair mask is a Schur multiplier in the row variables. These are independent statements. Low divisor rank does not remove an arbitrary joint row multiplier, and bounded entries do not imply bounded Schur or projective norm.

5 Determinant coefficients and unconditional frequency dispersion

We now retain the three cutoff-shell channels in their matched signed combination. Write

$$\mathcal{S}_{\mathrm{sh}}(\mathcal{E}) = \sum_{\alpha,\gamma,j} \mathfrak{a}^{\mathrm{sh}}_{\alpha,\gamma,j}\mathbf{1}_{\mathcal{E}(\alpha,\gamma,j)}, \tag{103}$$

where $\mathfrak{a}^{\mathrm{sh}}_{\alpha,\gamma,j}$ is the complete signed physical summand: it contains both opened-row coefficients, the actual row-pair mask, the fixed-period and smooth factors, and the matched kernel

$$\mathsf{A}_{m_\alpha,U_0}(j)\mathsf{A}_{m_\gamma,U_0}(j) - \mathsf{A}_{m_\alpha,T}(j)\mathsf{A}_{m_\gamma,T}(j).$$

Thus no triangle inequality separates the two mixed channels from the both-ultra channel in the definition below.

Put

$$c_{\alpha,\gamma}(j) = (m_\alpha j + h, m_\gamma j + h), \qquad \Delta^{\#}_{\alpha,\gamma}(j) = \frac{m_\alpha - m_\gamma}{c_{\alpha,\gamma}(j)}. \tag{104}$$

For $1 \le C \ll_{\mathscr{D}} Q$, define the signed normalized-determinant coefficient

$$A_C(n) := \mathcal{S}_{\mathrm{sh}}\big(c_{\alpha,\gamma}(j) \le C,\ \Delta^{\#}_{\alpha,\gamma}(j) = n\big), \qquad n \in \mathbb{Z}. \tag{105}$$

The off-diagonal row mask makes $A_C(0) = 0$. To avoid confusing this coefficient value with a Fourier frequency, we shall always write $\widehat{A}_{C,q}(0)$ for the latter.

The physical row interval has length $O_{\mathscr{D}}(Q)$. Fix a constant $B_{\mathscr{D}} \ge 1$ such that

$$\operatorname{supp} A_C \subset [-B_{\mathscr{D}}Q, B_{\mathscr{D}}Q] \cap (\mathbb{Z} \setminus \{0\}) \tag{106}$$

for every C. Indeed, $|\Delta^{\#}| = |m_\alpha - m_\gamma|/c \le B_{\mathscr{D}}Q$. In particular the diameter of this support, rather than merely its cardinality, is $O_{\mathscr{D}}(Q)$; this is the quantity entering the additive large sieve below.

5.1 From the inherited ℓ^1 and singleton bounds to energy

Let $A_C^+(n)$ be the absolute majorant formed by summing the absolute values of the complete matched physical summands in (105). The physical absolute-mass interface and the singleton version of the determinant-support theorem of TPC-31 give, for every fixed $\varepsilon > 0$,

$$\sum_n A_C^+(n) \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon N_0, \qquad N_0 := JQ^2 \asymp XQ, \tag{107}$$

$$\max_n A_C^+(n) \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon Q \log(2L)\{C + J\log(2C)\}. \tag{108}$$

For clarity, the second estimate follows by taking the allowed determinant set at each exact content to be the singleton $\{n\}$: its physical support count is at most one, and

$$\sum_{c \le C} \left(1 + \frac{J}{c}\right) \ll C + J\log(2C).$$

Both estimates remain valid for the matched sum because the absolute majorant of the sum of its three expanded channels is bounded by the sum of their TPC-31 majorants [12].

Proposition 5.1 (Determinant coefficient norms)**.** *For every $1 \le C \ll_{\mathscr{D}} Q$,*

$$\|A_C\|_1 \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon N_0, \tag{109}$$

$$\|A_C\|_\infty \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon Q(C+J), \tag{110}$$

$$\|A_C\|_2^2 \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon JQ^3(C+J). \tag{111}$$

If $C \le J$, then

$$\boxed{\|A_C\|_2^2 \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon \frac{N_0^2}{Q}.} \tag{112}$$

Proof. Since $|A_C(n)| \le A_C^+(n)$, (107) gives (109). Equation (108) gives (110) after enlarging the soft exponent to absorb $\log(2L)\log(2C)$. Apply both inherited estimates with $\varepsilon/2$ and use

$$\sum_n |A_C(n)|^2 \le \sum_n A_C^+(n)^2 \le \big(\max_n A_C^+(n)\big) \sum_n A_C^+(n).$$

This proves (111), after relabelling the soft exponent. If $C \le J$, then

$$JQ^3(C+J) \ll J^2Q^3 = \frac{(JQ^2)^2}{Q} = \frac{N_0^2}{Q},$$

which proves the last assertion. $\square$

The argument is deliberately sign-blind only at the final norm bound. The coefficient A_C itself was formed after the three physical channels were matched. Thus every future improvement of (112) coming from their signed interaction can be inserted without changing the Fourier statements below.

5.2 A no-wrap determinant Fourier transform

Choose an integer

$$Y > 2B_{\mathscr{D}}Q + 2.$$

Bertrand's postulate supplies, for all sufficiently large Q, a prime q with $Y < q < 2Y$. We fix Y to be the least integer above the displayed threshold. Consequently

$$q \asymp_{\mathscr{D}} Q, \qquad q > \operatorname{diam}(\operatorname{supp} A_C). \tag{113}$$

Define

$$\widehat{A}_{C,q}(r) := \sum_{n\in\mathbb{Z}} A_C(n)e_q(-rn), \qquad e_q(x) = \exp(2\pi i x/q), \qquad r \pmod q. \tag{114}$$

Theorem 5.2 (No-wrap Parseval and exceptional frequencies)**.** *Let q satisfy (113). Then*

$$\boxed{\frac{1}{q}\sum_{r\,(q)} |\widehat{A}_{C,q}(r)|^2 = \sum_{n\in\mathbb{Z}} |A_C(n)|^2.} \tag{115}$$

If $C \le J$ and $C \ll_{\mathscr{D}} Q$, then, for every $\sigma > 0$,

$$\boxed{\#\left\{r \pmod q : |\widehat{A}_{C,q}(r)| > N_0X^{-\sigma}\right\} \ll_{\varepsilon,h,\mathscr{D}} X^{2\sigma+\varepsilon}.} \tag{116}$$

The exceptional set in (116) is allowed to contain $r = 0$.

Proof. Ordinary finite Fourier Parseval gives

$$\frac{1}{q}\sum_{r\,(q)} |\widehat{A}_{C,q}(r)|^2 = \sum_{a\,(q)} \left|\sum_{n\equiv a\,(q)} A_C(n)\right|^2.$$

The diameter condition in (113) makes reduction modulo q injective on $\operatorname{supp} A_C$, so the last expression is exactly $\sum_n |A_C(n)|^2$. If E frequencies exceed the threshold in (116), then

$$EN_0^2X^{-2\sigma} \le \sum_{r\,(q)} |\widehat{A}_{C,q}(r)|^2 = q\|A_C\|_2^2 \ll X^{\varepsilon}\frac{q}{Q}N_0^2.$$

Now $q \asymp_{\mathscr{D}} Q$, and the result follows. ∎

The no-wrap choice is useful but not cosmetic. If one merely takes an arbitrary $q \asymp Q$, determinants separated by a multiple of q occupy the same residue cell, and the right side of Parseval is the energy of residue-cell sums rather than the coefficient energy itself.

5.3 Farey-family dispersion

For a reduced rational frequency a/v, put

$$\widehat{A}_C(a/v) := \sum_n A_C(n)e_v(-an). \tag{117}$$

The classical additive large sieve applies to an arbitrary complex coefficient sequence; no factorability hypothesis is needed here [4].

Theorem 5.3 (Additive large sieve for determinant frequencies)**.** *For $V \geq 2$, $C \leq J$, and $C \ll_{\mathscr{D}} Q$,*

$$\boxed{\sum_{2\leq v\leq V} \sideset{}{^*}\sum_{a\,(v)} |\widehat{A}_C(a/v)|^2 \ll_{\varepsilon,h,\mathscr{D}} X^{\varepsilon}(Q+V^2)\frac{N_0^2}{Q}.} \tag{118}$$

Consequently

$$\begin{aligned} &\#\left\{(a,v) : 2 \leq v \leq V, (a,v) = 1, |\widehat{A}_C(a/v)| > N_0 X^{-\sigma}\right\} \\ &\qquad \ll_{\varepsilon,h,\mathscr{D}} X^{2\sigma+\varepsilon}\left(1+\frac{V^2}{Q}\right). \end{aligned} \tag{119}$$

If $Q = X^{\beta+o(1)}$ and $V = X^{\rho+o(1)}$ with $\rho > 0$, the exceptional proportion among the $\asymp V^2$ reduced fractions is power-small for every fixed

$$\boxed{\sigma < \min\{\rho, \beta/2\}.} \tag{120}$$

Proof. Translate the integer support interval in (106) into an interval of positive integers. Translation changes each transform by a unit-modulus factor. The interval has length at most $2B_{\mathscr{D}}Q + 1$, so the additive large sieve gives

$$\sum_{2\leq v\leq V} \sideset{}{^*}\sum_{a\,(v)} |\widehat{A}_C(a/v)|^2 \ll_{\mathscr{D}} (Q+V^2)\|A_C\|_2^2.$$

Insert (112). Chebyshev's inequality gives (119).

The number of reduced fractions with denominator at most V is $\asymp V^2$. If $2\rho \leq \beta$, division of (119) by V^2 gives the exponent $2\sigma - 2\rho + o(1)$. If $2\rho \geq \beta$, it gives $2\sigma - \beta + o(1)$. Both are negative precisely in the strict range (120). □

Remark 5.4 (What the large sieve omits). The sum in (118) begins at $v = 2$, so all of its reduced frequencies are nonzero. One may add the single point $0/1$ to the large-sieve family, but the inequality then permits that one value to be as large as the natural scale N_0. A density-one conclusion for rational frequencies therefore says nothing about the distinguished coefficient $\widehat{A}_{C,q}(0)$ required by the untwisted physical reassembly. The next section isolates exactly the extra input needed there.

6 The distinguished zero frequency and a conditional transfer theorem

The notation in this section is connected to the canonical factorization by

$$\widehat{A}_{C,q}(r) = \widehat{\mathcal{S}}^{\mathrm{sh}}_{[1,C],q}(r), \qquad \mathcal{S}^{\mathrm{all}}_{\mathrm{sh}} = \mathcal{S}^{\mathrm{sh}}_{\mathrm{full}}, \qquad \mathcal{S}_{\mathrm{sh}}(c > C) = \mathcal{S}^{\mathrm{sh}}_{>C}. \tag{121}$$

Here $r = 0$ is the finite Fourier frequency of the normalized determinant. It is neither the orbit-variable Poisson zero mode nor the centered divisor-kernel zero mode used in earlier parts of the program.

The frequency $r = 0$ in (114) is independent of the auxiliary modulus:

$$\widehat{A}_{C,q}(0) = \sum_n A_C(n) = \mathcal{S}_{\mathrm{sh}}(c_{\alpha,\gamma}(j) \leq C). \tag{122}$$

It is therefore the untwisted small-content shell, not a typical oscillatory determinant twist. This section first records its exact relation to the complete raw shell and then states a quantitative additional hypothesis under which the TPC-30 large-content estimate closes the splice.

6.1 Equivalence to the full shell modulo large content

Let

$$\mathcal{S}_{\mathrm{sh}}^{\mathrm{all}} := \mathcal{S}_{\mathrm{sh}}(\text{all target contents}). \tag{123}$$

Since exact contents partition the physical triples,

$$\mathcal{S}_{\mathrm{sh}}^{\mathrm{all}} = \widehat{A}_{C,q}(0) + \mathcal{S}_{\mathrm{sh}}(c_{\alpha,\gamma}(j) > C). \tag{124}$$

TPC-30 proves, separately for the two mixed channels and the both-ultra channel, the large-content estimate

$$|\mathcal{S}_{\mathrm{sh}}(c > C)| \ll_{\varepsilon,h,\mathscr{D}} X^{\varepsilon}\left(Q^2 + \frac{XQ}{C}\right) \ll_{\mathscr{D}} X^{\varepsilon} N_0 \left(\frac{1}{J} + \frac{1}{C}\right), \tag{125}$$

where the matched version follows by adding those three bounds [11, 12].

Proposition 6.1 (Zero coefficient and full-shell equivalence)**.** *For every $1 \le C \ll_{\mathscr{D}} Q$ in the range of the inherited large-content theorem,*

$$\boxed{|\mathcal{S}_{\mathrm{sh}}^{\mathrm{all}} - \widehat{A}_{C,q}(0)| \ll_{\varepsilon,h,\mathscr{D}} X^{\varepsilon} N_0 \left(\frac{1}{J} + \frac{1}{C}\right).} \tag{126}$$

Consequently, an estimate for the zero coefficient transfers to the complete matched raw shell with precisely the two displayed losses, and an estimate for the complete raw shell transfers back to the zero coefficient with the same losses.

Proof. Subtract (122) from (124) and apply (125). The reverse transfer is the same inequality with the two terms interchanged. □

The content-projector factorization gives the coefficient-level version of this elementary partition. With λ_C defined in (75), proposition 3.11 states exactly that

$$\lambda_C(1) = 1, \qquad \sum_{\substack{t|g \\ t>1}} \lambda_C(t) = -\mathbf{1}_{g>C}.$$

Thus the principal $t = 1$ term is the untwisted full-content shell, while all nonprincipal projector terms reassemble exactly to minus the large-content shell. There is no unrecorded cancellation obtained by expanding the small-content indicator.

6.2 Zero-frequency flatness

The no-wrap Parseval identity identifies $\|A_C\|_2$ as the root-mean-square size of its determinant Fourier coefficients. The missing statement is that the distinguished zero coefficient is not much larger than this rms scale.

Definition 6.2 (Zero-frequency flatness exponent)**.** Suppose A_C is not identically zero. Its zero-frequency concentration ratio is

$$\mathfrak{F}_0(A_C) := \frac{|\sum_n A_C(n)|^2}{\sum_n |A_C(n)|^2} = \frac{|\widehat{A}_{C,q}(0)|^2}{\|A_C\|_2^2}. \tag{127}$$

We say that the packet has flatness exponent at most χ if

$$\boxed{\mathfrak{F}_0(A_C) \le X^{\chi+o(1)}.} \tag{128}$$

If $A_C \equiv 0$, we set $\mathfrak{F}_0(A_C) = 0$.

Condition (128) is an additional arithmetic input. It is not proved by Parseval, the additive large sieve, the matched rank-two identity, or any theorem in this paper. The notation "flatness" describes only the relative size of the distinguished Fourier coefficient; it is not a probabilistic assumption.

Theorem 6.3 (Conditional zero-frequency transfer)**.** *Assume*

$$Q = X^{\beta+o(1)}, \qquad J = X^{1-\beta+o(1)}, \qquad 0 < \beta < 1, \tag{129}$$

and choose

$$C = X^{\kappa+o(1)} \le J, \qquad C \ll_{\mathscr{D}} Q \tag{130}$$

for a fixed $\kappa \ge 0$. If the actual matched coefficient A_C satisfies (128) for a fixed χ, then

$$|\widehat{A}_{C,q}(0)| \ll X^{o(1)} N_0 X^{-(\beta-\chi)/2}, \tag{131}$$

$$|\mathcal{S}_{\mathrm{sh}}^{\mathrm{all}}| \ll X^{o(1)} N_0 \left\{ X^{-(\beta-\chi)/2} + X^{-(1-\beta)} + X^{-\kappa} \right\}. \tag{132}$$

In particular, if

$$\chi < \beta, \qquad \kappa > 0, \tag{133}$$

then every fixed

$$\boxed{\sigma < \min\left\{1-\beta,\ \kappa,\ \frac{\beta-\chi}{2}\right\}} \tag{134}$$

is a power-saving exponent relative to $N_0 \asymp XQ$ for the complete matched raw shell.

Proof. By the flatness hypothesis and proposition 5.1,

$$|\widehat{A}_{C,q}(0)| \le X^{\chi/2+o(1)} \|A_C\|_2 \ll X^{o(1)} N_0 X^{-\beta/2+\chi/2},$$

which is (131). Insert this estimate and the scale relations $J^{-1} = X^{-(1-\beta)+o(1)}$ and $C^{-1} = X^{-\kappa+o(1)}$ into (126). This proves (132). The strict range (134) absorbs all $o(1)$ and soft losses. □

Remark 6.4 (Conditional means conditional)*.* The conclusion of theorem 6.3 is a rigorous implication, but it is not an unconditional estimate for the physical shell. In particular, the fact that all but a power-small proportion of the nonzero frequencies obey the desired bound does not verify (128) at $r = 0$.

7 The high-β ledger and the $1/400$ gate

We now specialize the preceding unconditional and conditional statements to the high-row-scale packet inherited from TPC-28–TPC-31. Its exponents are

$$\boxed{\beta = \frac{267}{400}, \qquad 1-\beta = \frac{133}{400}.} \tag{135}$$

Thus

$$Q = X^{267/400+o(1)}, \qquad J = X^{133/400+o(1)}, \qquad N_0 = JQ^2 \asymp XQ. \tag{136}$$

All endpoint statements below retain the usual strict-margin convention: every fixed exponent below the displayed endpoint is proved, but equality at an endpoint is not asserted.

7.1 Exact fraction arithmetic

The small numerical gap driving this section is best kept in exact fractions:

$$1-\beta=\frac{133}{400}=\frac{266}{800}, \tag{137}$$

$$\frac{\beta}{2}=\frac{267}{800}, \tag{138}$$

$$\frac{\beta}{2}-(1-\beta)=\frac{267}{800}-\frac{266}{800}=\boxed{\frac{1}{800}}, \tag{139}$$

$$3\beta-2=\frac{801}{400}-\frac{800}{400}=\boxed{\frac{1}{400}}, \tag{140}$$

$$2(1-\beta)-\beta=\frac{266}{400}-\frac{267}{400}=-\boxed{\frac{1}{400}}. \tag{141}$$

The $1/800$ quantity is an amplitude-exponent gap. The $1/400$ quantity is its squared-energy counterpart and also the density exponent for the exceptional frequency set at the large-content target scale.

7.2 An unconditional density-one frequency theorem

Take

$$C\le J \tag{142}$$

and choose the no-wrap prime $q\asymp Q$ from theorem 5.2. Then

$$\left(\frac{1}{q}\sum_{r\,(q)}|\widehat{A}_{C,q}(r)|^2\right)^{1/2}\ll X^{o(1)}N_0X^{-267/800}. \tag{143}$$

Thus the rms determinant frequency has a saving larger than the TPC-30 large-content margin by exactly $1/800$ in the exponent.

Corollary 7.1 (High-β almost-all-frequency saving)**.** *Fix $0<\eta<133/400$ and put*

$$\sigma_\eta=\frac{133}{400}-\eta. \tag{144}$$

For every fixed $\varepsilon>0$,

$$\#\left\{r\pmod q:|\widehat{A}_{C,q}(r)|>N_0X^{-133/400+\eta}\right\}\\ \ll_{\varepsilon,h,\mathscr{D}}X^{266/400-2\eta+\varepsilon}, \tag{145}$$

$$\frac{1}{q}\#\left\{r\pmod q:|\widehat{A}_{C,q}(r)|>N_0X^{-133/400+\eta}\right\}\\ \ll_{\varepsilon,h,\mathscr{D}}X^{-1/400-2\eta+\varepsilon+o(1)}. \tag{146}$$

In particular, after choosing $\varepsilon<\eta$, a proportion $1-O(X^{-1/400-\eta+o(1)})$ of the frequencies has the stated saving. The exceptional set may contain $r=0$.

Proof. Insert (144) into (116):

$$2\sigma_\eta=\frac{266}{400}-2\eta.$$

This proves (145). Since $q=X^{267/400+o(1)}$, division by q and (141) give (146). $\square$

The conclusion is stronger than merely saying that a generic frequency has some power saving. It says that the precise saving required to meet the 133/400 large-content floor already holds outside a set of relative density $X^{-1/400+o(1)}$. It does not identify the distinguished zero frequency with the density-one set.

The Farey-family version has the same rms endpoint $\beta/2$. Taking $V = X^{267/800+o(1)} = Q^{1/2+o(1)}$ in theorem 5.3 gives, outside a power-small proportion of reduced rational frequencies, every fixed saving below

$$\min\left\{\frac{267}{800}, \frac{267}{800}\right\} = \frac{267}{800}. \tag{147}$$

Again, this family contains no copy of the zero rational frequency because its denominators begin at two.

7.3 The exact conditional threshold

For the full-shell splice choose

$$C = \lfloor J \rfloor = X^{133/400+o(1)}, \qquad \kappa = \frac{133}{400}. \tag{148}$$

The two large-content losses in (132) then both have endpoint 133/400. Because the conclusion asks for every strict saving below this endpoint, the flatness term reaches it precisely when

$$\begin{aligned} \frac{\beta-\chi}{2} &\geq \frac{133}{400} \\ &\iff \chi \leq \beta - \frac{266}{400} = \frac{267}{400} - \frac{266}{400} = \boxed{\frac{1}{400}}. \end{aligned} \tag{149}$$

Corollary 7.2 (Conditional high-β closure of the matched shell)**.** *Assume that the actual matched determinant coefficient with the choice* (148) *satisfies*

$$\mathfrak{F}_0(A_C) \leq X^{\chi+o(1)} \qquad \textit{for some fixed } \chi \leq \frac{1}{400}. \tag{150}$$

Then every fixed

$$\boxed{\sigma < \frac{133}{400}} \tag{151}$$

is a power-saving exponent relative to $N_0 \asymp XQ$ for the complete matched raw cutoff shell.

Proof. Under (150), (149) shows that the flatness margin is at least 133/400. The other two entries in (134) both equal 133/400 by (148). Apply theorem 6.3 with a strict final exponent. □

Condition (150) has a transparent amplitude interpretation. By definition,

$$|\widehat{A}_{C,q}(0)| \leq X^{\chi/2+o(1)}\|A_C\|_2. \tag{152}$$

Thus the boundary $\chi \leq 1/400$ permits the zero coefficient to be larger than the rms determinant coefficient by a factor at most

$$X^{1/800+o(1)}. \tag{153}$$

At $\chi = 0$, the zero frequency is no larger than the rms scale up to $X^{o(1)}$, and its saving endpoint is 267/800; the final splice is still limited to $266/800 = 133/400$ by large content. By contrast, the unconditional Cauchy bound permits $\mathfrak{F}_0(A_C) \asymp Q = X^{267/400+o(1)}$, which gives no saving at all. Moving from that worst case to (150) is the genuine arithmetic task; the small numerical slack does not make it a soft consequence of the frequency average.

Quantity	Exact exponent	Status
Large-content splice margin $1-\beta$	$133/400 = 266/800$	Unconditional
Rms determinant-frequency margin $\beta/2$	$267/800$	Unconditional
Rms amplitude slack over the splice	$1/800$	Exact arithmetic
Bad-frequency density exponent at the splice target	$1/400$	Unconditional
Allowed squared-flatness loss	$\chi \leq 1/400$	Additional input
Complete matched-shell saving	every $\sigma < 133/400$	Conditional on flatness

The last row concerns only the matched raw cutoff shell delivered to this paper. Even under the flatness input it does not, by itself, prove a prime-pair asymptotic, positivity, twin primes, or a breach of the sieve parity barrier.

8 Sharp interface obstructions and a primitive witness

The preceding mean-square estimate is nearly sufficient numerically, but it cannot select the coefficient $r = 0$. We give two exact models showing that this failure is structural at the current interface rather than an artifact of a loose inequality.

8.1 A flat determinant signal

Proposition 8.1 (Perfect nonzero-frequency cancellation does not control zero)**.** *Let q be prime and let I consist of q consecutive integers. Define*

$$A(n) = \frac{N_0}{q}\mathbf{1}_{n\in I}. \tag{154}$$

Then

$$\|A\|_1 = N_0, \qquad \|A\|_\infty = \frac{N_0}{q}, \qquad \|A\|_2^2 = \frac{N_0^2}{q}, \tag{155}$$

but its q-point determinant transform satisfies

$$|\widehat{A}(0)| = N_0, \qquad \widehat{A}(r) = 0 \quad (1 \leq r < q). \tag{156}$$

Proof. The norm formulas are immediate. The consecutive integers in I form a complete residue system modulo q, so

$$\widehat{A}(r) = \frac{N_0}{q} e_q(-rn_0) \sum_{a=0}^{q-1} e_q(-ra).$$

The geometric sum is q for $r = 0$ and zero otherwise. □

Taking $q \asymp Q$, (155) saturates the ℓ^1, ℓ^∞, and ℓ^2 scales available for A_C. Thus those three norms, the support length, and even exact vanishing of every nonzero Fourier coefficient cannot imply any positive flatness saving at zero. This is an abstract interface countermodel; it is not a lower bound for the physical Möbius shell.

8.2 The coherent mode of the content projector

For positive integers $N_1, \dots, N_M$, define

$$P_C(i,j) = \mathbf{1}_{(N_i,N_j)\le C}. \tag{157}$$

The content-projector identity gives the exact signed rank-one expansion

$$P_C = \sum_{t\ge 1} \lambda_C(t) v_t v_t^{\mathsf{T}}, \qquad v_t(i) = \mathbf{1}_{t|N_i}. \tag{158}$$

The coefficients are signed, so this is not a positive-semidefinite decomposition.

Proposition 8.2 (A coherent constant-row mode)**.** *If the N_i are pairwise coprime and $N_i > C$ for every i, then*

$$P_C = \mathbf{J}_M - I_M. \tag{159}$$

Its spectrum is

$$M-1, \qquad -1 \textit{ with multiplicity } M-1, \tag{160}$$

and

$$\|P_C\|_{2\to 2} = M-1, \qquad \operatorname{srank}(P_C) := \frac{\|P_C\|_F^2}{\|P_C\|_{2\to 2}^2} = \frac{M}{M-1}. \tag{161}$$

Proof. For $i \ne j$, pairwise coprimality gives $(N_i, N_j) = 1 \le C$; on the diagonal $(N_i, N_i) = N_i > C$. This proves (159). The all-ones vector has eigenvalue $M-1$, and its orthogonal complement has eigenvalue -1. Finally, $\|P_C\|_F^2 = M(M-1)$, giving (161). □

The stable rank being close to one is not a gain here: it records that one coherent direction dominates the operator. In particular, restricting to small canonical content is not a row-centering operation.

8.3 A finite primitive realization

The coherent model occurs inside the actual arithmetic interface. Take

$$h = 2, \quad j = 1, \quad d = 1, \quad R = 12, \quad T = 50, \quad U_0 = 200, \quad C = 30, \tag{162}$$

and the prime rows

$$59, 71, 101, 107, 137, 149, 179, 191, 197. \tag{163}$$

Their targets $m+2$ are respectively

$$61, 73, 103, 109, 139, 151, 181, 193, 199, \tag{164}$$

which are distinct primes exceeding C. Hence the 9×9 content matrix is $\mathbf{J}_9 - I_9$, with spectrum $8, -1, \dots, -1$ and stable rank $9/8$.

For completeness, the finite Ramanujan-model divisor coefficient is

$$\lambda'_R(u) = u \sum_{b \le R/u} \frac{\mu(ub)\mu(b)}{\varphi(ub)} \qquad (u \le R). \tag{165}$$

At $R = 12$,

$$\lambda'_{12}(1) = \sum_{b\le 12} \frac{\mu(b)^2}{\varphi(b)} = \frac{113}{30}. \tag{166}$$

Since each target in (164) is prime and lies in $(T, U_0]$, its prefix and ultra increment are

$$\mathsf{A}_{m,50} = -\frac{113}{30}, \qquad C_m = \log(m+2). \tag{167}$$

For two distinct witness rows with targets p_i, p_j, the sum of the three raw channels is therefore the formal prime-log polynomial

$$\boxed{\log p_i \log p_j - \frac{113}{30}(\log p_i + \log p_j).} \tag{168}$$

Its coefficient of $\log p_i \log p_j$ is one, so there is no universal algebraic annihilation among the three channels.

This finite realization certifies the algebraic interface only. Although the displayed rows happen to be members of a finite list of prime pairs at gap two, their presence proves neither a growing lower bound nor any asymptotic statement about twin primes. It also does not prove that the physical shell has a coherent main term; cancellation from the remaining row coefficients and phases is still possible.

8.4 The arbitrary-mask obstruction

Low rank in divisor variables does not survive an uncontrolled conclusion about the row operator. Given any finite kernel $K_{\alpha,\gamma}$, an entrywise bounded mask may be chosen to align its phases, for example

$$\mathfrak{m}(\alpha,\gamma) = \begin{cases} \overline{K_{\alpha,\gamma}}/|K_{\alpha,\gamma}|, & \alpha \neq \gamma \text{ and } K_{\alpha,\gamma} \neq 0, \\ 0, & \alpha = \gamma \text{ or } K_{\alpha,\gamma} = 0. \end{cases} \tag{169}$$

Then the masked off-diagonal scalar sum is $\sum_{\alpha\neq\gamma} |K_{\alpha,\gamma}|$. Therefore no theorem uniform over all entrywise bounded pair masks can deduce signed cancellation from the matched rank-two identity alone. The actual mask may have additional structure, but that structure must be stated and quantified rather than inferred from (101).

9 Technology comparison and the next affine–Möbius gate

The exact identities above expose several familiar analytic forms, but none of the cited technologies accepts the complete physical coefficient without an additional argument. We record the boundary to prevent a formal rearrangement from being mistaken for a cancellation theorem.

9.1 One Möbius sign versus two affine signs

Classical uniform bounds of Davenport type estimate a single oscillatory Möbius sum

$$\sum_{d \asymp D} \mu(d)\omega(d)e(\theta d) \tag{170}$$

with arbitrary logarithmic saving when D is a polynomial scale and ω has controlled smoothness; see, for example, Iwaniec and Kowalski [4, Chapter 13]. Fixed-period characters and Mellin phases can be included without changing the essential one-variable form. This is useful for a bare row coefficient, but opening a reflected ultra leg changes the arithmetic problem.

Fix ℓ, j and a reflected divisor s on primitive support. The condition

$$s \mid \ell dj + h \tag{171}$$

has a unique solution $d \equiv d_0 \pmod{s}$, because $(s, \ell j) = 1$. Write $d = d_0 + st$. The complementary quotient is

$$U(t) = \frac{\ell j(d_0 + st) + h}{s} = u_\star + \ell jt, \qquad u_\star = \frac{\ell j d_0 + h}{s}, \tag{172}$$

and the two affine forms satisfy the nondegeneracy identity

$$\boxed{su_\star - \ell j d_0 = h \neq 0.} \tag{173}$$

After the exact sign fusion, a typical opened coefficient contains

$$\mu(d_0 + st)\mu(u_\star + \ell jt). \tag{174}$$

This is a two-point affine Möbius correlation, not the one-sign sum (170).

Logarithmically averaged two-point Chowla–Elliott theorems control fixed nonproportional affine forms in an averaged regime [6]. The physical packet here asks additionally for uniformity as s, ℓ, j and the affine coefficients grow polynomially, for dyadic rather than logarithmic averaging, for fixed residue factors, and for reassembly over all rows and cutoffs. We do not claim that the existing theorem supplies this package.

9.2 Spectral and Kloosterman technology

Bilinear and trilinear estimates for Kloosterman fractions can yield strong cancellation once the variables, moduli, smooth weights, and coefficient norms match the hypotheses of the relevant theorem [1, 2]. Modern exceptional spectral large sieves provide further gains in carefully structured families [5]. In the present shell, however, the modulus is tied to reflected target divisors, the prime–Möbius coefficient depends on a moving factorization, and the generic pair mask is still joint. No parameter ledger in this paper verifies the hypotheses of one of those stronger estimates. They are possible technologies for a future refined packet, not black boxes used in our theorems.

9.3 A concrete next target

The conditional transfer theorem reduces a high-β closure of the raw shell to the exact boundary $\chi \leq 1/400$. A robust route with a fixed margin is to prove, for the actual physical amplitude and $C \asymp J$, a power-saving estimate of the form

$$\left|\sum_n A_C(n)\right|^2 \ll X^{1/400-\eta} \sum_n |A_C(n)|^2 \tag{175}$$

for some fixed $\eta > 0$. A viable proof would have to exploit information absent from the abstract norm interface, such as:

(a) cancellation in the affine correlation (174) uniform over a controlled subfamily;

(b) a projective or operator-norm decomposition of the actual pair mask;

(c) an additional centering identity that annihilates the coherent determinant mode before absolute reassembly; or

(d) a decomposition into spectral/Kloosterman ranges with a complete conductor and norm ledger.

Each option can be tested independently. Failure of one does not refute the others, and a sharp impossibility result for any proposed interface would itself refine the route map.

For the full calibrated cutoff difference, the two drift legs also impose the inherited saving L^{-1}; hence any final exponent must be intersected with the fixed prime-source margin. Even a proof of (175) would close only this residual packet. It would not by itself supply positivity, undo all earlier reductions, or prove a prime-pair theorem.

9.4 Scope of the advance

The advance in this paper is that the determinant-frequency route no longer ends at the vague instruction "control the zero mode." It proves that ordinary determinant frequencies already have the required size outside a power-small exceptional set, identifies the remaining hard raw component of the original calibrated residual with one special member of that set, quantifies the exact additional flatness needed, and shows that the current Hilbert-space data cannot force it. The remaining door is narrow, but it is still an open analytic estimate rather than a claimed consequence of the existing large sieve.

10 Exact certificate and scope ledger

The companion script

`experiments/tpc32_certificate.py`

is a standard-library Python regression certificate for the finite algebra and rational bookkeeping used in this paper. Run from the paper directory with Python 3.10 or later:

```
python experiments/tpc32_certificate.py
python -O experiments/tpc32_certificate.py
```

Both modes use exception-based checks rather than `assert`, perform exactly

$$\boxed{488{,}845} \tag{176}$$

recorded checks, and regenerate byte-identical JSON.

The check families are:

Finite check family	Recorded checks
Matched shell, rank at most two, and polarization	4,017
Exact content inversion and the λ_C projector	410,980
Canonical phase splitting, including $(c, q) > 1$	22,850
Primitive prime–Möbius sign fusion	22,254
Exact Parseval and the flat-signal zero-frequency model	8,490
ℓ^2-versus-$\ell^1\ell^\infty$ inequality	19,530
High-β rational ledger and the 1/400 threshold	20
Coherent matrices and the primitive finite witness	566
$\lambda'_{12}(1)$ and formal prime-log polynomials	110
Scope flags	28
Total	488,845

The certificate uses integers and `fractions.Fraction`. Finite character orthogonality is represented in exact cyclotomic coordinates, not by floating-point roots of unity. In particular, it checks:

(a) the scalar and matrix matched-shell identities, symmetric polarization, and rank bound;

(b) exact gcd inversion, the cumulative content projector, and $\lambda_C(t) = 0$ for $2 \le t \le C$ and $\lambda_C(t) = -1$ for $C < t \le 2C$;

(c) the canonical phase lift to modulus cq in cases with $(c, q) > 1$, without introducing an inverse of c mod q;

(d) primitive target-divisor coprimality and both prime–Möbius fusion identities;

(e) no-wrap character orthogonality, exact Parseval, and a constant signal whose nonzero Fourier coefficients all vanish;

(f) the exact high-β identities

$$1 - \frac{267}{400} = \frac{133}{400}, \qquad 3\frac{267}{400} - 2 = \frac{1}{400};$$

(g) the spectrum and stable rank of $\mathbf{J}_M - I_M$, including the nine-row primitive realization;

(h) $\lambda'_{12}(1) = 113/30$ and the nonzero formal coefficient of $\log p_i \log p_j$ in the three-channel witness.

Reproducibility values are:

normalized certificate-script source SHA-256	`9ccb7e40d31b4bd23b79df09336fcee6` `230d95c7a9f1a85da0324fb6eb807d1e`, (177)
JSON SHA-256	`77ac56e2f4f3543224876e8a0374564` `c81c1b4a9f17800f18ca5249e12954d36`, (178)
certificate digest	`a890734f5450cda4a7a400f5e1d63796` `83eb4b9e53a05c8497f256edfdc82c78`. (179)

10.1 Claim boundary

The unconditional mathematical statements established in this paper are:

(i) the exact matched cutoff shell, its rank-two divisor tensor, and its calibrated drift boundary;

(ii) exact canonical-content inversion, determinant-phase splitting, and the retained-multiplier bilinear representation;

(iii) the prime–Möbius one-leg fusion and the explicit separation boundary for a generic pair mask;

(iv) determinant ℓ^1, ℓ^∞, and ℓ^2 bounds inherited from the physical singleton-cell theorem;

(v) no-wrap Parseval, exceptional-frequency compression, and the classical Farey large-sieve estimate;

(vi) the exact equivalence of the auxiliary zero coefficient and the full matched raw shell modulo the previously bounded large-content contribution;

(vii) the high-β almost-all-frequency proportion and the two sharp finite interface obstructions.

The flatness transfer theorem is a proved implication whose premise

$$|\widehat{A}_{C,q}(0)|^2 \leq X^{\chi + o(1)} \sum_n |A_C(n)|^2$$

is explicitly additional. The paper does not prove this premise, does not show that $r = 0$ is a typical determinant frequency, and does not assert that the generic pair mask has bounded projective norm.

Neither the certificate nor the analytic theorems prove:

(i) signed cancellation of the complete Möbius-tail residual;

(ii) an improved level of distribution for primes;

(iii) a Hardy–Littlewood prime-pair asymptotic or positivity statement;

(iv) the twin-prime conjecture; or

(v) a breach of the sieve parity barrier.

The finite prime-row witness is an exact algebraic regression test only. The negative JSON scope flags make any future promotion of that witness, or of the conditional 1/400 gate, into an unconditional prime-pair claim detectable during review.

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